

DAYLIGHT

Lightweight brightening day cream

A premium, warm-climate O/W prototype engineered for immediate comfort, elegant slip and responsible long-term cosmetic support.



Development brief

A 100 g bench prototype for India. Non-steroidal, hydroquinone-free, no medical claims. Designed around one shared pH window and a deliberately short active stack.

Master-data lock: formula hash 9dd84162fc57 • 19 line items • 100.00 g total • prepared 02 October 2026

1. What success looks like

The fastest realistic wins are sensory and optical: a smoother water-rich surface, less visible flaking, soft slip and a healthier-looking sheen. Tone and blemish support are slower biological outcomes—not four-day promises.

Window	What a user may notice	What not to promise
First use	Comfortable hydration, softer slip, less tightness; temporary glow from a smoother hydrated surface.	Permanent glow or pigment removal.
24 hours	More supple feel; dry patches may look less obvious.	Acne cure or guaranteed brightening.
2–4 days	Better repeat-use feel; less dull-looking surface for some users.	Guaranteed spot fading.
1 week	More consistent softness and comfort if used daily.	Clinical-level tone change.
2–4 weeks	Barrier comfort and even-looking tone may begin to look more consistent.	Melasma treatment.
6–12 weeks	Best window to assess tone support; sunscreen use is essential.	Drug-like acne or pigmentation claims.

Use context: apply a pea-sized amount to clean, dry skin in the morning, then broad-spectrum SPF 30+ sunscreen. Sunscreen is the non-negotiable partner for any tone-evening goal.

2. Final 100 g master formula

This exact table is the single source used to generate the companion PowerPoint. pH-adjuster quantity is a planning quantity; in a real batch it is q.s. to pH 5.0–5.5, with any difference compensated by water.

Ingredient	INCI	ml	g	Phase	Purpose
Distilled / deionised water	Aqua	65.25	65.25	A	Solvent / continuous phase
Disodium EDTA	Disodium EDTA	0.10	0.10	A	Chelator
Glycerin, vegetable IP/USP	Glycerin	4.00	4.00	A	Humectant
Xanthan gum, clear grade	Xanthan Gum	0.20	0.20	A	Rheology / stabiliser
Niacinamide	Niacinamide	4.00	4.00	A	Barrier, tone and sebum support
Allantoin	Allantoin	0.20	0.20	A	Soothing / smoothing
Cetearyl olivate & sorbitan olivate (Olivem 1000)	Cetearyl Olivates, Sorbitan Olivates	4.00	4.00	B	Primary O/W emulsifier
Cetearyl alcohol	Cetearyl Alcohol	0.50	0.50	B	Co-emulsifier / body
Caprylic/capric triglyceride	Caprylic/Capric Triglyceride	4.00	4.00	B	Light emollient
Squalane, olive or sugarcane	Squalane	2.00	2.00	B	Light emollient / barrier lipid
Tocopherol, mixed	Tocopherol	0.20	0.20	B	Oil antioxidant
Reserved distilled / deionised water	Aqua	10.00	10.00	C	Cool-down solvent
Sodium hyaluronate, cosmetic grade	Sodium Hyaluronate	0.10	0.10	C	Humectant / film former
Alpha-arbutin, >=99% with COA	Alpha-Arbutin	1.50	1.50	C	Tone-evening active
Zinc PCA, cosmetic grade	Zinc PCA	0.50	0.50	C	Sebum / NMF support
D-panthenol 75% liquid	Panthenol	2.00	2.00	C	Humectant / soothing
Phenoxyethanol & ethylhexylglycerin (PE 9010 type)	Phenoxyethanol, Ethylhexylglycerin	1.00	1.00	C	Broad-spectrum preservative
IFRA-certified facial fragrance	Parfum	0.15	0.15	C	Sensory
10% lactic acid solution, q.s.	Lactic Acid, Aqua	0.30	0.30	D	pH adjustment

3. Why this formula feels premium

Premium feel comes from the architecture—not from a long ingredient list. A non-ionic emulsifier, light esters, a little squalane and controlled rheology create a short, silky break with a soft finish.

Ingredient	Texture contribution	Skin feel	User experience
Olivem 1000 + cetearyl alcohol	Fine O/W structure and controlled body	Cushion without wax drag	Cream looks rich but spreads quickly.
CCT + squalane	Light emollient slip	Dry-touch softness	Less greasy than heavy oils; a healthy glow.
Glycerin + panthenol + sodium hyaluronate	Layered humectancy and light film	Comfort and plumpness	Hydrated, not wet or sticky when balanced.
0.20% xanthan	Low-level structure	Smooth, stable pickup	No watery separation; not a gel.
0.15% IFRA fragrance	Low fragrance load	Clean, modern sensory cue	Noticeable but not loud; omit if sensitive.

The professional trick is restraint: too much glycerin feels tacky, too much fatty alcohol feels waxy, too much oil feels slow, and too much fragrance raises irritation risk. This formula keeps each lever modest.

4. Ingredient lessons in plain language

Ingredient family	What it does	Why it is here	When benefit can show
Humectants: glycerin, panthenol, sodium hyaluronate	Hold water near the skin surface.	Fast hydration and comfortable finish.	Minutes to days.
Niacinamide 4%	Supports barrier, tone evenness and sebum balance.	Broadest evidence-to-risk fit for this brief.	Comfort early; tone usually weeks.
Alpha-arbutin 1.5%	Helps slow a pigment-forming pathway.	Adds a second, complementary tone lever.	Assess 6–12 weeks, not 4 days.
Zinc PCA 0.5%	Mineral salt with modest sebum/NMF support.	Helps keep oily/combination skin in scope.	Subtle; weeks.
Allantoin 0.2%	Soothing and smoothing.	Improves comfort and rough-feel perception.	Days.
Emollients	Reduce roughness by filling gaps between skin flakes.	Creates the “wow” slip and soft finish.	Immediate.
EDTA + preservative	Control metal-driven drift and microbes.	Protect safety and stability.	Invisible but essential.

5. Beginner manufacturing recipe — exactly 100 g

Work in a clean, ventilated area. This is a cosmetic prototype, not a sterile preparation. Use a calibrated 0.01 g scale and record every lot number.

STEP 1 — Equipment & sanitization

What to do: Gather two 250 mL borosilicate beakers, 0.01 g balance, probe thermometer, mini stick blender or frother, spatulas, water bath, pH meter with pH 4.01/7.00 buffers, gloves, mask, hair covering, 70% IPA, sanitised airless pump. Wash with hot detergent, rinse with distilled water, then spray/submerge in 70% IPA for 5 minutes and air-dry. **Time:** 15–20 min. **See:** clean dry tools. **Avoid:** using wet IPA or touching sanitised parts.

STEP 2 — Weighing

What to do: Tare a weigh boat and weigh every line item from the master table. Pre-weigh Phase A, B and C separately. Reserve the 10.00 g water for cool-down. Prepare 10% lactic acid only if needed. **Avoid:** measuring 0.10 g ingredients with kitchen spoons.

STEP 3 — PHASE A: water phase

What to do: Slurry 4.00 g glycerin with 0.20 g xanthan until smooth. Into Beaker A add 65.25 g water, 0.10 g EDTA, 4.00 g niacinamide, then 0.20 g allantoin; stir 1 minute after each. Add the slurry. Heat in a water bath to 70–75°C and hold 15–20 min with gentle stirring. **See:** uniform, slightly viscous liquid. **Avoid:** dumping xanthan directly into water—it fish-eyes.

STEP 4 — PHASE B: oil phase

What to do: Add 4.00 g CCT, 2.00 g squalane, 4.00 g Olivem 1000 and 0.50 g cetearyl alcohol. Heat to 70–75°C until every flake is melted; hold 5–10 min. Add 0.20 g tocopherol for the final minute. **See:** clear homogeneous oil. **Avoid:** combining while flakes remain.

STEP 5 — EMULSIFY

What to do: Bring A and B within 3–5°C. Pour B slowly into A over 20–30 seconds. High-shear mix with the head submerged in 20–30 second bursts for 2–3 minutes; then stir gently while cooling. **Total:** continue gentle mixing until below 45°C. **See:** opaque uniform cream, no oil beads. **Avoid:** whipping air or stopping all mixing while it cools.

6. Cool-down, pH, filling

STEP 6 — Prepare cool-down solution

What to do: At the start, sprinkle 0.10 g sodium hyaluronate over 10.00 g reserved water and hydrate 20–30 min. Below 45°C, dissolve 1.50 g alpha-arbutin and 0.50 g zinc PCA into this gel. Add to the emulsion and mix 1–2 min. Below 40°C add 2.00 g panthenol and mix 1 min. **See:** no visible powder or stringy clumps. **Avoid:** heating alpha-arbutin.

STEP 7 — Preservative & fragrance

What to do: Below 40°C add 1.00 g PE 9010-type preservative and mix 2 min. Below 35°C add 0.15 g facial fragrance and mix 1–2 min. For fragrance-free, omit fragrance and replace it with 0.15 g distilled water. **Avoid:** using essential oils or candle/soap fragrance.

STEP 8 — pH

Target: pH 5.0–5.5 at 25°C. Calibrate meter with pH 4.01 and 7.00 buffers. Measure neat with a suitable cosmetic probe, or 1 g cream + 9 g water as a recorded 1:10 screening dilution. To lower, use 10% lactic acid solution, 2–3 drops at a time, stir 60 sec, re-check. To raise, use 10% sodium citrate or L-arginine solution. Never add neat NaOH at home. **Avoid:** chasing a number with large additions.

STEP 9 — Final mix & fill

What to do: Mix gently 2–3 min below 35°C. It should be smooth, opaque, uniform and non-gritty. Fill below 30°C into a sanitised opaque 30–50 g airless pump using a syringe or piping bag; minimise headspace. Rest 24–48 h before judging final viscosity. **Avoid:** a wide-mouth jar or fingers.

STEP 10 — Storage

Store sealed, dark and below 30°C; avoid bathroom heat and sunlight. For an untested home prototype, label “LAB SAMPLE — NOT FOR SALE” and use a conservative 4–6 week development window pending microbial and stability testing. Do not infer shelf life from appearance alone.

7. Preservation, safety and claims

Water-based creams can look perfect while carrying bacteria, yeast or mould. Raw powders, water, tools, hands and repeated jar opening are all contamination routes. PE 9010-type preservative at 1.0% is selected for its broad utility and pH tolerance, but only a preservative challenge test demonstrates protection.

Important distinctions: tocopherol protects oils from oxidation; it does not preserve water. Essential oils are not a complete preservative system and can add allergens. “Natural” does not mean safer. Do not use crushed medicines, hydroquinone, steroids, pharmacy azelaic-acid creams or unknown extracts as cosmetic raw materials.

Patch test: apply a small amount to the inner forearm or behind ear for 24–48 h, then cautiously introduce to face. Stop for persistent burning, swelling, hives or worsening rash. Never claim “side-effect free.” Pregnancy, eczema, rosacea or active acne concerns merit clinician input.

Commercial gate: before sale in India, use documented cosmetic raw materials, batch records, GMP manufacture, stability, preservative efficacy/challenge, microbial limits, packaging compatibility, safety assessment, compliant label and applicable Cosmetics Rules/BIS requirements. A home batch is not commercially released product.

8. India sourcing & Ahmedabad development leads

Prices are indicative planning ranges, not quotations. Verify current stock, MOQ, COA, SDS, GST/shipping and lot traceability directly. Ask specifically for cosmetic grade and a certificate of analysis.

Need	Specification / pack	Planning range	Potential source / note
Small-pack actives	Niacinamide, alpha-arbutin, panthenol, sodium hyaluronate, Zinc PCA, allantoin; 10–100 g	■190–850 per small pack	India Beauty Raw Material; Cosmesi Global; Purenso Select; Moksha Lifestyle; TRCKem.
Base materials	Olivem 1000, CCT, squalane, cetearyl alcohol, xanthan, EDTA; 50–500 g	■100–1,200 per pack	Cosmetic raw-material retailers; request COA/SDS.
Preservative	PE 9010 type, 100 mL	■180–400	Ases / specialist cosmetic ingredient retailers; verify ratio and INCI.
Fragrance	Facial leave-on, IFRA Category 5B + allergen declaration, 10–30 mL	■200–900	Moksha Aromatics, Keva / S H Kelkar / specialist fragrance houses.
Equipment	0.01 g scale, pH meter, beakers, probe, mixer, airless pumps	■3,000–8,000	Scientific suppliers, Borosil, IndiaMART, packaging vendors.

Ahmedabad / Gujarat leads to screen: KRIYAM Therapeutics (Ahmedabad; formulation development, private label, manufacturing) — <https://www.kriyam.in/> • Aelicare Wellness (Ahmedabad; custom formulation, small-batch capability claimed) — <https://aelicarewellness.com/> • Cosmenova (Changodar, Ahmedabad; concept-to-shelf claims) — <https://www.cosmenova.in/> • Zymo Cosmetics (Changodar; custom/private label; verify MOQ) — <https://zymocosmetics.com/> • Orchid Lifesciences (Ahmedabad/Vatva listings; verify current R&D; scope) • Aelicare/HCP Wellness for Gujarat-scale-up conversations. These web pages are leads, not endorsements; suitability for a 100 g bench sample must be confirmed.

Ask before sharing the formula: Will you sign an NDA? Do you make 100–500 g lab samples? Who owns the formula and revisions? Can you provide COA/SDS and INCI review? What are development fees, number of iterations, MOQ, lead time, stability/microbial testing scope, fragrance/IP terms, batch records and regulatory support?

9. Cost planning & testing

Indicative material economics

Scale	Raw material use (planning)	Packaging	Interpretation
100 g prototype	~■90–■250 consumed; small-pack purchase outlay commonly ■3,000–■7,000	■80–■200 airless pump	Purchase outlay is high because you buy small packs.
500 g bench batch	~■450–■1,250 consumed	■400–■1,000 for 5–10 units	Bulk buys reduce unit cost.
1 kg pilot	~■750–■2,000 consumed at small-B2B rates	■800–■2,000	Excludes labour, freight, labels and rejects.
Professional development	—	—	~■15,000–■75,000+ depending on iterations, IP and lab work; request written quote.
Testing	—	—	Microbiology ■1,000–■4,000; stability/compatibility/challenge often ■10,000–■50,000+; lab-dependent.

Testing plan

Test	Home screen?	Professional requirement
Appearance, odour, texture, pH, colour	Yes; record day 0, 1, 7, 14, 28.	Instrumental colour/viscosity if commercial.
Freeze/thaw, 4°C / room / 40°C observation	Screen only; useful for red flags, not proof.	Formal stability protocol and interpretation.
Package compatibility / pump function	Yes with filled packs.	Longer-term compatibility and transport simulation.
Patch test	Cautious self-screen only.	Dermatological safety / HRIPT as appropriate.
Microbial count & preservative challenge	No.	Cosmetic microbiology lab; challenge test is essential for launch.

10. Prototype learning loop

A — baseline: assess after 48 h for slip, tack, absorption, pH and separation. **B — too sticky:** reduce glycerin 0.5–1%, increase water; do not immediately add more oil. **C — too greasy:** reduce squalane or CCT by 0.5–1% and replace with water. **D — too thin:** confirm 24–48 h maturation, then test xanthan +0.05% or cetearyl alcohol +0.2% in a controlled split batch. **E — pilling:** reduce total film-formers, check application amount and rub time, and screen under sunscreen. **F — fragrance:** compare 0, 0.05 and 0.15% while keeping all else fixed. **G — stability:** do not “fix” separation by adding random emulsifier; recheck phase temperatures, emulsifier melt and shear.

Professional developers change one or two variables at a time, retain a control, score sensory attributes blind, and keep a batch record. Never assume the first formula is final.

11. Formula guardrails & references

Compatibility guardrails: keep finished pH 5.0–5.5; add alpha-arbutin and sodium hyaluronate below 40°C; add preservative below 40°C and fragrance below 35°C; avoid strong acids, pure L-ascorbic acid, retinoids and drug actives in this day-cream jar; keep Zinc PCA away from high-pH carbomer systems.

Fragrance-free option: remove Parfum 0.15% and use 0.15 g additional Aqua. Master formula above remains the fragranced version; any fragrance-free pilot is a controlled variant, not a silent change.

References and live verification links

- Cosmetic Ingredient Review (CIR), Niacinamide safety assessments — <https://www.cir-safety.org/>
- European Commission SCCS, Opinion on alpha-arbutin (face cream safety context) — https://health.ec.europa.eu/scientific-committees/scientific-committee-consumer-safety-sccs_en
- EU CosIng ingredient database — https://single-market-economy.ec.europa.eu/sectors/cosmetics/cosmetic-products-database_en
- India Cosmetics Rules, 2020 / CDSCO cosmetics portal — <https://cdsco.gov.in/>
- India Beauty Raw Material — <https://indiabeautyrawmaterial.com/>
- KRIYAM Therapeutics — <https://www.kriyam.in/>
- Aelisure Wellness — <https://aelisurewellness.com/>
- Cosmenova — <https://www.cosmenova.in/>
- Web search leads for Gujarat manufacturers were checked 02 October 2026; confirm all claims, facilities and quotes directly before engagement.